John Benjamin Cassel

Current Direction

Discovery-driven domain design and software engineering of engineering design, operational planning, automated equipment, and research support platforms for stewarding natural and built infrastructure by multiple criteria.

Work experience

◇ **Principal Software Engineer** (Fall 2026-), Terion (remote)

- Uncovering new ways to leverage science, data, and AI to build agricultural resilience.

◇ **Principal Software Engineer** (Summer 2024-Summer 2026), Regrow Agriculture (remote)

- Fostered a consolidation that integrated disparate MRV services into a whole maintainable by a single agent-empowered team, leading to continued increase in shipped per-engineer value-weighted functionality.
- Established systems guaranteeing the completeness and consistency of program submission data.
- Introduced workflows for packaging the outcomes of sustainability programs, automating much manual effort.
- Co-developed a new product offering with AI features using existing data assets to open a new growth area.
- Supported leadership, project managers, and engineering managers on roadmap planning, cross-team collaboration, proposal development, review, mentoring, triage, workshop moderation, and vision leadership.

◇ **Staff Software Engineer/Principal Software Engineer** (Spring 2022-Summer 2022/Summer 2022-Spring 2024), Indigo Agriculture (remote)

- led a company-scale initiative successfully guiding multiple teams to synthesize multiple kinds of integrated data into an integrated schema capable of serving a variety of sustainability programs, including data sourced from remote sensing, farm equipment, sensors, and partner integrations, allowing significant consolidation.
- Designed and co-developed a sensor logistics system while maintaining the existing solution.
- Staff/Principal duties similar to Regrow Agriculture position.

◇ **Research Programmer** (Spring 2018-Spring 2022), Wolfram|Alpha LLC (Champaign, IL)

- Formulated and undertook a biology content gathering effort to help win a major partner.
- Led the biomolecular sequence vertical for the Wolfram Language.

◇ **Systems Engineer; Tech Lead** (Summer 2015-Winter 2017), Agrible (Champaign, IL)

- Co-created the main task processing system of the analysis backend.
- Codeveloped the primary representation and interface for farm activities.
- Served as the technical lead of the team providing cross-product superstructure.

◇ **Research Programmer** (Spring 2014-Summer 2015), Wolfram Research (Champaign, IL)

- Created an engineering design framework supporting agent-based exploration.
- Assisted in the development of semantic import functionality.
- Supported Wolfram|Alpha database infrastructure.

◇ **Software Engineer** (Fall 2009-Winter 2013), Wolfram|Alpha LLC, (remote/Champaign, IL)

- Enhanced an internal object-relational toolkit with inheritance and geo-temporal support.
- Developed and maintained W|A database release infrastructure.
- Advised multiple object-relational schema designs.

◇ **Member of the Technical Staff** (Summer 2008-Fall 2009) Wolfram Research, (Champaign, IL)

- Engineered a database release system with full-deployment and fast-patching features.
- Developed runtime access libraries for database data.

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- Wrote developer and curator tools for data management, including a version control system.
- Designed machine learning tools for anticipating user behavior.

◇ **Research Engineer** (Fall 2005 – Summer 2008), Riverglass Incorporated (Champaign, IL)

- Devised a domain-specific knowledge resource editor with rich constraint checking.
- Invented a new probabilistic network technique for modeling intelligence scanning tasks.

◇ **Research Consultant** (Spring 2005 – Fall 2005), Riverglass Incorporated (Champaign, IL)

- Constructed a knowledge-base with probabilistic inference rules.
- Built a planning system for evaluating the importance of analytics tasks.

Education

◇ **OCAD University**, Toronto, ON, Canada
M.Des. in Strategic Foresight and Innovation, May 2011
Major Project: *Addressing Risk Governance Deficits through Scenario Modeling Practices.*
Advisers: Peter Jones with Walter Derzko
Committee review: John's work as demonstrated in the MRP can be recognized as an important contribution to systemic foresight theory and practice. ...It has a serious moral thrust in its ability to deal effectively with problems of significant scale and complexity. Because of this temper, this methodology can ...facilitate breakthroughs of understanding, consensus for action, and the coordination of social power.

◇ **University of Illinois**, Urbana-Champaign, IL
B.Sc. in Computer Science with Honors, May 2002.
Application Sequence: *Manufacturing Engineering.*

Skills

◇ **Processes and Domains** Experience with discovery-based design processes, domain-driven design, software engineering, geographic information systems, data processing pipelines, Agile practices, the data science process, testing strategy, strategic foresight, and agentic workflows. Familiarity with agricultural operations, factory simulation, reliability and quality control, basic bioinformatics, and cloud/distributed/parallel computing.

◇ **Analytical Techniques** Stakeholder analysis, non-parametric Bayesian inference, decision-theoretic planning, simulation, optimization, knowledge representation and reasoning, machine learning emphasizing reinforcement learning, neural networks, system dynamics modeling, domain-specific languages, and data visualization.

◇ **Programming Languages** Professional experience in Python, Java, SQL, Common Lisp, and Wolfram Language (Mathematica). Brief experience in many others, including Rust, Go, Erlang, C++, C, and Ruby.

◇ **Tools, Libraries, and Frameworks** Django, MySQL, PostgreSQL, Claude Code, Snowflake, AWS, GCP, Docker, Ruby on Rails, Wolfram SystemModeler, ArcMap, Tableau, Terraform, FastAPI, Spring, and others.

Publications

◇ **2014** Cassel, J. Probabilistic Programming with Stochastic Memoization: Implementing non-parametric bayesian inference. *Mathematica Journal*, 16:1.

◇ **2014** Cassel, J. Non-parametric stakeholder discovery: A process for mitigating risk governance deficits through open-ended protocols. In Hsu, W. H. (editor) *Emerging Methods in Predictive Analytics: Risk Management and Decision-Making*, pages 97-126. IGI Global, Hershey, PA.

◇ **2014** Cassel, J. The Methodological Unboundedness of Limited Discovery Processes. *FORMacademisk, 7:4.*

◇ **2016** Cassel, J. Wolfram|Alpha: A Computational Knowledge "Search" Engine. In Lee, N. (editor) *Google It: Total Information Awareness*, pages 267-299. Springer, New York, NY.

◇ **2018** Cassel J.B., Cousineau S.V. Permaculture as a Systemic Design Practice. In: Jones P., Kijima K. (editors) *Systemic Design. Translational Systems Sciences, vol 8., pages 293-318. Springer, Tokyo.*

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